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Automatic Restoration and Analysis of Birchbark Manuscripts

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1. Introduction

The computational study of historical written documents sits at the intersection of epigraphy, computer vision, and computational linguistics. The subject of the present work is the Novgorodian birchbark manuscripts — documents scratched on pieces of birchbark that have been discovered, and continue to be discovered, in the vicinity of Veliky Novgorod and Staraya Russa, as well as a number of other locations across medieval Rus'.

The majority of manuscripts is written in the Old Novgorodian dialect (OND), which differs substantially from both standard Old Russian and Church Slavonic. The corpus of direct OND sources is considerably small. It includes, in addition to the birchbark manuscripts themselves, a number of epigraphic inscriptions on objects, some marginal notes in ecclesiastical books, and, with certain qualifications, a portion of Novgorodian parchment and paper charters in which dialectal features predominate over supradialectal norms. Indirect sources — oriented primarily towards Church Slavonic norms but containing Novgorodian features — encompass the remaining Novgorodian parchment and paper charters, the Novgorodian chronicles (especially the First Novgorodian Chronicle), and ecclesiastical books (Zaliznyak, 2004: 11–13).

Many birchbark manuscripts have survived only in fragmentary condition, making their restoration and interpretation a considerable challenge. Computational models can serve as valuable assistants in this process, providing additional evidence to support and complement the work of human scholars. The aim of the present work is to develop a pipeline for filling lacunae in birchbark manuscripts written in OND, as well as for the temporal dating and categorical attribution of manuscripts. The small size of the corpus and the distinctive features of the dialect make this task particularly challenging.

In a longer-term perspective, the text restoration module described here could be integrated into a broader birchbark manuscript processing pipeline, also encompassing optical character recognition from photographs and the automatic segmentation of continuous handwritten text into words.

2. Literature Review

2.1 Birchbark Manuscripts and Related Work

Before turning to the literature on automatic ancient text restoration, it is important to survey work that engages directly with the birchbark manuscript corpus or with the broader domain of Old Russian and Church Slavonic. Computational approaches to handwritten historical documents have grown considerably in recent years, with contributions covering optical character recognition, script identification, handwriting analysis, and authorship attribution. However, studies addressing Cyrillic sources remain comparatively rare, and tend to

focus on documents from the 18th century onwards (Diao et al., 2025: 18–19). The works discussed below are therefore particularly noteworthy.

2.1.1 Script Recognition and Palaeographic Dating

Sidorov (2018) proposed a method for the automatic palaeographic dating of birchbark manuscripts based on letterform features. The system combines classical appearance modelling with convolutional neural networks, and outputs predictions via an ensemble of 171 support vector machines — one for each interval between the identified chronological periods. The method achieved a mean absolute error of 18.9 years, a figure the author reports as comparable to the accuracy of expert palaeographers.

Malashin et al. (2025) addressed the complementary task of character recognition directly on birchbark images. Analysing the manuscript database, the authors identified the 33 most frequent letters and generated a large synthetic training dataset by rendering these characters in varied scripts with added noise to simulate the surface texture of real birchbark. Several YOLO (You Only Look Once) object detection models were trained and evaluated; following hyperparameter optimisation, YOLOv10 achieved the best result with a mean Average Precision (mAP) of 0.89. The authors identify the high level of noise and physical damage inherent to birchbark, as well as the inconsistent and sometimes extremely thin stroke widths, as the principal challenges of the task.

Although character recognition is an important prerequisite for a fully automated pipeline, it should be noted that for the existing transcribed corpus, the more pressing challenge lies not in reading individual letters but in interpreting damaged passages and recovering lost text. The present work focuses precisely on this task.

2.1.2 Textual and Dialectal Analysis

Afanasev and Lyashevskaya (2025) applied corpus-based language distance measurement to study the diatopic variation of the Old Novgorodian dialect. Using FastText embeddings trained on character n-grams (length 3–6), which allow robust operation on noisy data and small corpora without the biases associated with pre-trained transformers, the authors conducted analysis at the level of 3-character shingles. UPGMA clustering confirmed the chronological heterogeneity of the dialect and identified three stable periods — 1020–1160, 1160–1220, and 1280–1430 — within each of which the manuscripts show a high degree of linguistic homogeneity.

Lendvai et al. (2023) adapted BERT for the classification of pre-modern Slavic texts along three parameters: manuscript identity, century of composition, and region of origin. Their approach uses domain-adaptive pre-training via unsupervised MLM fine-tuning on a historical

corpus spanning nearly a thousand years. The results confirm that domain adaptation improves classification accuracy, enabling the model to capture subtle dialectal and chronological distinctions even from relatively small datasets.

Lendvai et al. (2025) investigated automatic retrieval and alignment of parallel text passages across chronological and regional variants of Church Slavonic. The study compares multilingual semantic embeddings (LaBSE, SBERT) against character-level alignment algorithms under conditions of strong orthographic variation and OCR noise. Multilingual models such as LaBSE achieve high precision without domain-specific adaptation, though their performance is critically dependent on sentence segmentation quality. The Smith–Waterman algorithm proved competitive with neural approaches on heavily damaged or noisy texts.

2.2 Word Segmentation in Ancient Texts

Word segmentation — the task of identifying token boundaries in continuous text — is a prerequisite for a complete manuscript processing pipeline, since many historical documents, including most birchbark manuscripts, follow a *scriptio continua* convention without spaces between words. However, this task is typically not addressed in restoration-focused studies, which generally assume pre-tokenised, space-delimited input.

Robert-Hayek and Eyharabide (2025) proposed a method for the automatic segmentation of ancient Greek and Latin texts written in *scriptio continua*, based on hierarchical transformers. Their two-level architecture uses a character-level model to predict word boundaries and a word-level model to evaluate the linguistic plausibility of the resulting sequences, refined via self-training. The system achieves an F1 score of up to 96%, demonstrating that hierarchical modelling — simultaneously attending to graphemic features and global semantic context — outperforms standard transformer-based approaches on this task. While the full segmentation pipeline is out of scope for the present project, these results inform the longer-term integration perspective described in the Introduction.

2.3 Masked Language Modelling for Ancient Text Restoration

Approaches to lacuna restoration can broadly be divided into two categories: fine-tuned masked language models, which adapt pre-trained BERT-like architectures, and task-specific models trained from scratch, which are designed and optimised specifically for ancient text restoration.

2.3.1 Fine-tuned Masked Language Models

Following the introduction of BERT, researchers observed that the MLM objective used for its pre-training is structurally identical to the task of restoring lacunae in historical texts: in both cases, a model must predict a missing token given its surrounding context. This observation

has since motivated a growing body of work applying or adapting BERT-like models to ancient and medieval corpora.

Lazar et al. (2021) were among the first to exploit this parallel systematically, applying it to the restoration of lacunae in transliterated Akkadian texts from the Oracc corpus. During preprocessing, editorial uncertainty markers and alternative readings were removed, and texts were provided with word-space delimiters. In training, 15% of tokens were masked at random. Each missing token — a single Akkadian logographic sign — was represented by the special symbol 'x'; because Akkadian signs may correspond to multiple subword tokens in BERT's vocabulary, the authors adapted the k-beam algorithm to produce multi-subtoken predictions. Of the models compared, multilingual mBERT fine-tuned on Akkadian achieved the strongest performance, with Hit@5 of 89% and Mean Reciprocal Rank (MRR) of 83%. Notably, basic mBERT outperformed model pre-trained from scratch on Akkadian data, suggesting that broad multilingual pre-training may compensate for the absence of domain-specific pre-training.

Baidya et al. (2025) applied a similar framework to Sanskrit, a low-resource ancient language with a substantially larger corpus than OND. Their dataset comprised over one million sentences; they used a standard BERT architecture with a custom tokeniser employing Byte Pair Encoding (BPE) to handle Sanskrit's complex morphology and Devanagari script. The model achieved Hit@10 of 70%, outperforming SanBERTa (60%), a pre-existing Sanskrit language model, with a Character Error Rate of 23%.

Gu et al. (2024) proposed a more architecturally innovative approach targeting classical Chinese — a language written without word boundaries. The authors constructed two evaluation datasets: a short-text set (300 documents, segmented into 30–60 character fragments, yielding approximately 424,000 training and 47,000 test examples with single-word masking), and a long-text set (400 documents, segmented into 256–512 character fragments, with variable masking rates of 10%, 20%, 30%, and 40% to assess model performance under progressively sparser context). Their model ACTRM extends SikuRoBERTa — a pre-trained model for classical Chinese — with a phrasal embedding that aggregates features from neighbouring tokens, designed to compensate for the absence of spaces. The standard MLP prediction head is replaced with a Kolmogorov–Arnold Network (KAN), which offers more parameter-efficient approximation and trainable activation functions. ACTRM maintains its advantage over competitors across all masking rates on the long-text dataset, with the performance gap widening as the proportion of masked tokens grows: Hit@5 declines from 76% at a 10% masking rate to 61.5% at 40%, with a corresponding increase in perplexity from 9% to 24%.

2.3.2 Task-specific Models Trained from Scratch

The approaches discussed above predict one token at a time. An alternative line of work instead predicts the entire missing sequence in a single pass.

Assael et al. (2019) introduced Pythia, the first deep learning model for ancient text restoration, applied to ancient Greek epigraphy. The model was trained on PHI-ML, a cleaned and machine-actionable version of the PHI corpus comprising over 3.2 million words across approximately 35,000 inscriptions spanning the 7th century BCE to the 5th century CE. Missing characters are represented by '-' and characters to be predicted by '?'; the target sequence is sampled from a contiguous span of up to ten characters within a broader context window. The architecture is a character-level sequence-to-sequence model with a bidirectional LSTM encoder and an attention-based LSTM decoder. To better capture word-level context the encoder concatenates character embeddings with embeddings of the words they belong to; partially damaged or out-of-vocabulary words are mapped to a shared unknown embedding. Predictions are returned as a ranked list of top-20 hypotheses via beam search. The best-performing variant achieved a character error rate of 30.1%, substantially outperforming expert epigraphists (57.3% CER), and the ground-truth restoration appeared within the top-20 hypotheses in 73.5% of cases. The authors additionally demonstrated that predictive performance increases monotonically with context length, peaking around 500 characters.

Building on Pythia, the same research group subsequently developed Ithaca (Assael et al., 2022), extending the framework in several important directions. Where Pythia focused solely on text restoration, Ithaca tackled three tasks simultaneously: textual restoration, geographical attribution, and chronological attribution. The model was trained on a substantially expanded dataset of approximately 3.8 million words across 78,000 inscriptions from the PHI corpus spanning the 7th century BCE to the 5th century CE. Architecturally, Ithaca replaces the LSTM encoder-decoder with a transformer torso consisting of eight stacked blocks with sparse multi-head attention, enabling more efficient handling of long sequences. As in Pythia, inputs combine character- and word-level representations, with damaged or unknown words rendered as '[unk]'. Physical damage was simulated using a geometrically distributed masking scheme with parameter $P = 0.1$ at the character level, and editorial restorations enclosed in square brackets were retained in the training data to maximise the amount of available text. Ithaca achieved a CER of 26.3% and a top-20 accuracy of 78.3%, substantially outperforming both Pythia and the expert historian baseline (59.6% CER). For geographical attribution, the model reached 70.8% top-1 and 82.1% top-3 accuracy across 84 regions. For chronological attribution, Ithaca's predictions fell within an average of 29.3 years of the ground-truth date intervals, with a median distance of only 3 years.

The successor system Aeneas (Assael et al., 2025) extended this framework to Latin epigraphy, training on over 176,000 inscriptions from three epigraphic databases covering approximately 16 million characters. Architectural innovations include the replacement of the BigBird-based torso with a T5 transformer decoder augmented with Rotary Position Embeddings (RoPE), and the removal of the word-level representation layer, with the model operating exclusively at the character level. A ResNet-based visual module is integrated for the geographical attribution task. Aeneas is also the first model in this line of work to handle gaps of unknown length, denoted by a ‘#’ symbol, using an auxiliary prediction head and a non-sequential beam search that at each step either extends the gap by one character or closes it. On the Latin epigraphic test set, Aeneas achieved a CER of 40.5% with known restoration length and 66.1% for gaps of unknown length, alongside a top-20 accuracy of 46.5%. Geographical attribution reached 72.3% top-1 accuracy across 62 Roman provinces, and chronological attribution achieved an average distance of 13 years from ground-truth date ranges, with a median of zero years.

Both the Ithaca and Aeneas papers confirm the combination of model predictions and expert scholar judgment consistently outperformed either working in isolation, establishing a compelling case for human–AI collaboration in the digital humanities.

3. Data

3.1 Dataset Sources

The dataset used for model training comprises nearly 3.8 million word tokens and was assembled from a variety of sources, subsequently filtered and normalised (see Section 3.2). It can broadly be divided into two parts: texts that are related to the Old Novgorodian dialect to varying degrees, and all remaining texts. Each of these parts is discussed in more detail below.

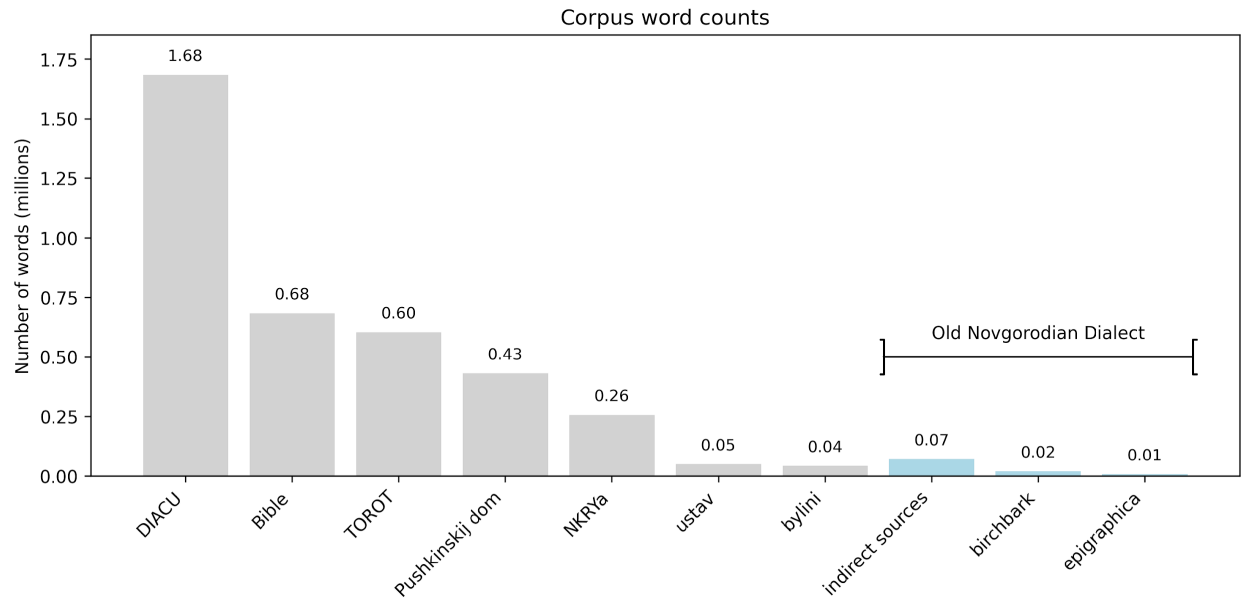


Fig. 1. Dataset statistics

3.1.1 Old Novgorodian Dialect

The open corpus of birchbark manuscripts, hosted at the portal Drevnerusskie berestyanye gramoty (gramoty.ru, accessed on 07.04.2026), currently contains 1,258 birchbark and 2 lead documents totalling around 24,000 words, with a vocabulary exceeding 4,100 lexical items. The birchbark manuscripts are short texts, with a median length of 65 characters or 12 words per document (Fig. 1). In practice, very short documents tend to carry too little contextual information to be useful for training or evaluation. We therefore retain only those documents exceeding 25 characters in length, yielding a working subset of 958 manuscripts with a total of over 21,000 words.

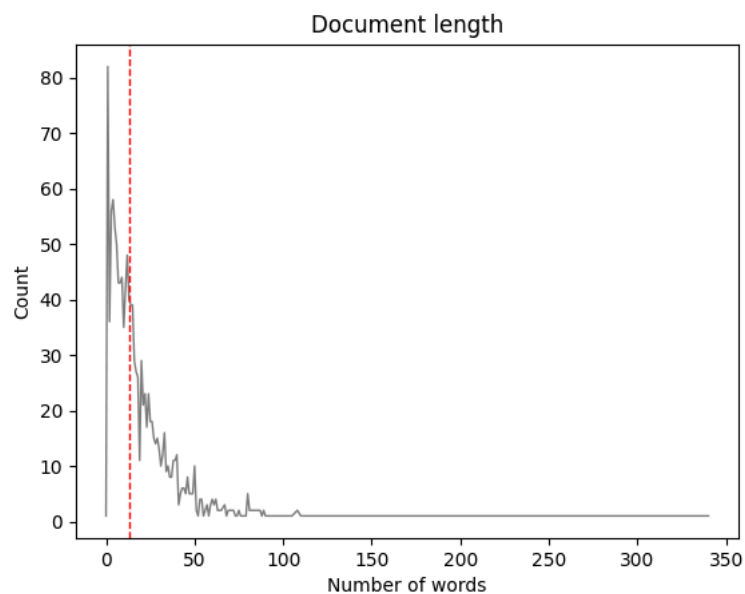


Fig. 2. Distribution of birchbark manuscripts' length and its median value

In the dataset, each document is provided in two forms — in the original writing without spaces (*scriptio continua*) and with word boundaries marked by spaces. Reflecting the layout of the documents, texts are divided into lines and contain hyphenation marks. The texts also employ a system of editorial conventions: scribal errors are enclosed in curly braces {}, partial reconstructions of letters based on surviving traces are marked with square brackets [], full reconstructions supported only by contextual inference are indicated by round brackets (), a single hyphen - denotes an unrestorable gap of one character, and an ellipsis ... denotes an unrestorable gap of unknown length.

The majority of birchbark manuscripts are damaged to some degree: more than two thirds of the full corpus (879 documents) contain gaps that are either irrecoverable or have not yet been reconstructed. Only 258 manuscripts, totalling over 5,000 words, are either fully intact or carry only partial reconstructions.

As shown in Figs. 2–3, the majority of documents originate from Novgorod (1,145 items) and the nearby town of Staraya Russa (58), with the remainder distributed across Torzhok, Smolensk, Pskov, Tver, Moscow, Zvenigorod Galitsky, Pereyasavl Ryazansky (Ryazan), Staraya Ryazan, Vologda, Vitebsk, and Mstislavl. Most manuscripts from Novgorod, Staraya Russa, Torzhok, and Pskov constitute the main direct source of the Old Novgorodian dialect in a broad sense (Zaliznyak, 2004: 6–7).

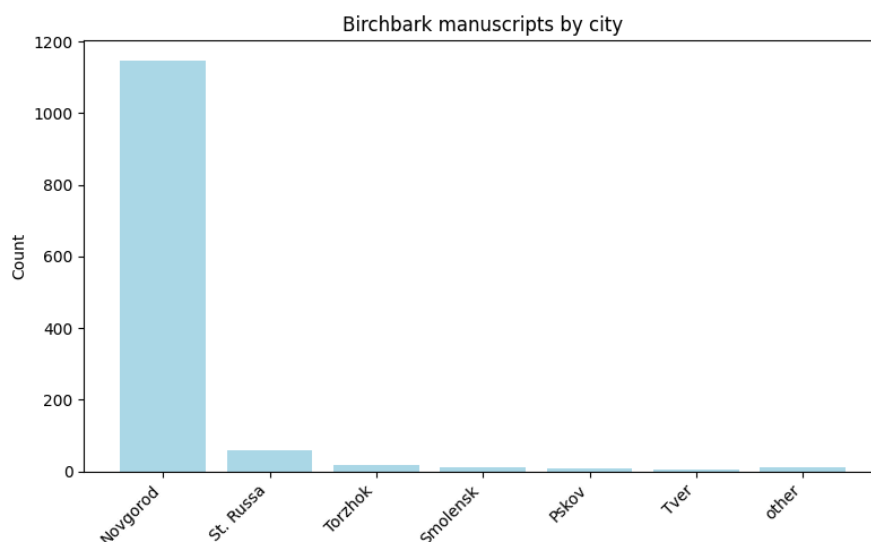


Fig. 3. Number of birchbark manuscripts by city



Fig. 4. Visualization of manuscripts' origin on map

As additional sources of the Old Novgorodian dialect, we use Novgorod (34,300 words) and Pskov (1,300) parchment charters and the First Novgorodian Chronicle (nearly 34,000 words). These documents differ stylistically from most birchbark manuscripts, yet retain a number of OND features. The Old Russian epigraphic dataset hosted at the portal Drevnerusskaja epigraphika (epigraphica.ru, accessed 07.04.2026) is also a structurally and partly thematically comparable source: its editorial conventions follow the same system as those used in the birchbark corpus. However, epigraphic inscriptions conform to Old Novgorodian norms only to a limited degree, as the majority of wall inscriptions follow Church Slavonic conventions. Applying the same filtering procedure as for the epigraphic manuscripts, we reduced the combined set from 10,200 to 7,100 words.

3.1.2 Other Sources

Given the comparatively limited amount of data available in the OND subset, it was decided to compile a broader dataset drawing on documents written in other Old Russian dialects, Church Slavonic, and a number of closely related languages. For this purpose, both existing datasets (DIACU, TOROT, and a biblical corpus of the Ponomar Project) and independently crawled or compiled materials were used. For a more detailed description of all datasets, see Appendix 1.

3.2 Birchbark Manuscripts' Metadata

Each birchbark manuscript in the dataset is accompanied by metadata. Of the available metadata, we use the *conventional date* (*uslovnaya data*) and the genre category as target variables for the auxiliary classifiers described in Section 4.3.

The conventional date, as defined by (Yanin & Zaliznyak, 2000: 136), is assigned by mapping each manuscript onto one of the segments of a chronological scale spanning the second quarter of the 11th century to the mid-15th century. The scale is divided primarily into 20-year segments (with 25-year segments for the 11th century and occasional 10-year segments for the 15th century). When the underlying stratigraphic interval is wide (40–60 years or more), the midpoint is taken as the basis for assignment, shifted towards any archaeologically preferred sub-interval where one exists; palaeographic, graphemic, or linguistic features may provide additional refinement. For a smaller number of manuscripts whose content permits content-based dating — through mentions of specific persons or events — the conventional date follows content rather than stratigraphy, and this applies even to manuscripts that lack a stratigraphic date altogether. The conventional date thus takes the form of a 20-year interval rather than a single year, reflecting the inherent uncertainty of archaeological dating.

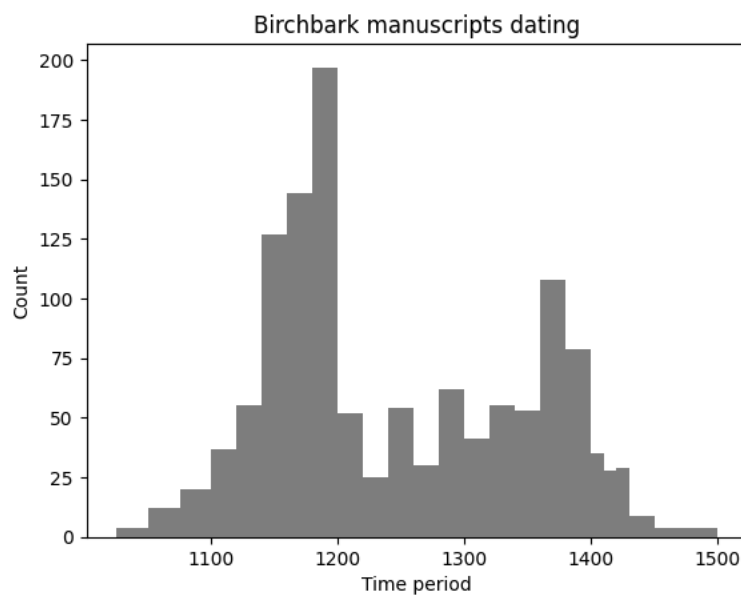


Fig. 5. Birchbark manuscripts conventional dating

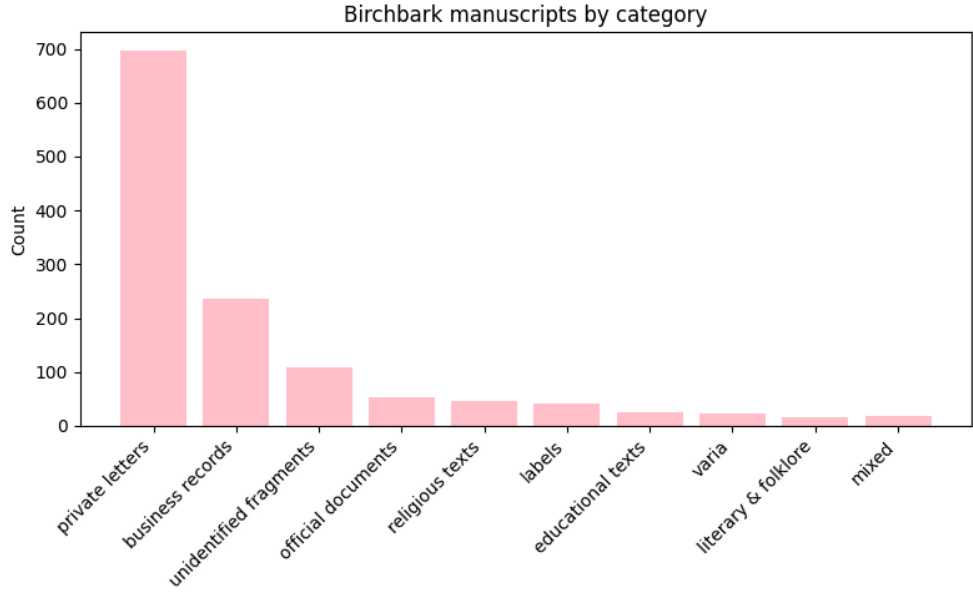


Fig. 6. Number of birchbark manuscripts by category (genre)

As shown in Figures 5–6, both the dating distribution and the category distribution are highly skewed. The majority of manuscripts cluster around the 12th and 14th centuries, while the dominant category is private letters, accounting for approximately 700 manuscripts.

For the dating prediction task, we discretise the timeline into nine bins of 50 years each, spanning 1050–1499. Four manuscripts dated 1025–1050 fall outside this range; as their interval overlaps with the first bin, they are assigned to it based on the overlap between their date interval and the bin boundary.

For the category prediction task, we merge the original fine-grained categories into four broader classes: *letters* (private letters), *records* (business records and official documents), *religious* (religious texts), and *other* (educational texts, folklore, varia, and content of uncertain nature). Mixed-category records (annotated with //) are excluded. Even after this consolidation, the class distribution remains imbalanced: letters — 696, records — 330, other — 171, religious — 45. This imbalance is addressed during training via weighted cross-entropy loss.

3.3 Text Preprocessing and Normalisation

During dataset processing, overlapping portions were removed to prevent data leakage: for instance, the birchbark manuscript corpus was excluded entirely from TOROT. A substantial number of documents from DIACU were also discarded due to poor OCR quality in the original dataset.

When processing birchbark manuscripts and epigraphic texts, the contents of curly braces {} were removed to prevent the model from training on ungrammatical forms, while the contents of both square brackets [] and round brackets () were retained to provide the model with additional context (for a detailed description of the notation system, see Section 3.1.1). Line break markers are removed, and hyphenated word fragments split across lines are rejoined into a single token.

In certain other source documents, angle brackets <> were used for editorial annotations; their contents were likewise removed. Other editorial abbreviations and marks not present in the original text were also stripped during preprocessing.

All text was lowercased. Of the available punctuation, three symbols were retained: the middle dot ·, the colon :, and the cross +, the latter having been used during certain periods to mark the beginning of a birchbark document (Zaliznyak, 2004: 37). Some rare characters are deleted, some transformed to their invariants (see Appendix 2 for normalisation scheme).

For models that require it, an additional normalisation step is applied to ensure compatibility with the model's built-in tokenizer (see Section 4.1).

3.4 Masking Scheme and Set Construction

Data are split into training, validation, and test sets in a 90/5/5 ratio. The test set is organised into two evaluation tracks: Test A for artificially generated gaps, and Test B for real lacunae based on editorial reconstructions.

Most birchbark manuscripts fit comfortably within BERT's maximum input length of 512 tokens. However, some documents — most notably those from other datasets — considerably exceed this limit. Long texts are therefore split into overlapping chunks of 256 tokens.

All sources are partitioned into training, validation, and test subsets in a 90/5/5 ratio. The test set is organised into two evaluation tracks. *Test A* is constructed by applying the masking procedure described below to fully intact text fragments, creating a benchmark with known ground-truth answers. *Test B* consists of documents from the birchbark and epigraphic corpora that contain at least one editorial reconstruction enclosed in square or round brackets: editorial reconstructions enclosed in brackets are concealed, and the model is evaluated on its ability to reproduce them character by character. Evaluating on two separate tracks allows a distinction to be drawn between the model's general restoration capacity — measured on Test A — and its practical utility for manuscript scholarship — measured on Test B.

Most birchbark letters fit comfortably within the maximum input length of 512 tokens. Documents that exceed this limit — primarily those drawn from larger chronicle sources — are split into non-overlapping chunks of 256 tokens.

The masking probability of 8% is set to match the empirically observed reconstruction density in the birchbark corpus. In Test B, 7,130 characters out of 96,606 are subject to editorial reconstruction, yielding a reconstruction rate of 7.38%. Including the 1,120 gap tokens already present in the same texts raises the combined ratio of masked and unreadable material to 8.54%. An 8% masking rate therefore reflects the proportion of damaged content in the target domain rather than an arbitrary hyperparameter choice. This value is held constant across both modelling approaches, so that any difference in restoration quality can be attributed to architectural factors rather than to differences in training conditions.

For BERT-based models, masking is applied using a span-based scheme with span lengths drawn from a geometric distribution with parameter $p = 0.35$. This produces predominantly short spans with a long tail, approximating the distribution of contiguous damage in historical documents. The overall masking budget is kept at 8% per sequence. The gap token, which marks genuinely unreadable positions already present in the training text, is excluded from the set of eligible mask targets: the model observes it as context but is never trained to predict it, preventing the model from learning to output [GAP] as a restoration.

Models trained from scratch use a purpose-built collator that simulates physical document degradation through three complementary strategies. *Span masking* expands randomly selected positions into contiguous runs, with span lengths sampled uniformly from $[1, max_span]$ (*max_span* is set to 3), reflecting the fact that physical damage tends to affect adjacent characters rather than isolated ones. *Edge masking* (applied with probability 0.10) selects a short span at the beginning or end of the sequence and masks it entirely, simulating the frayed or broken edges characteristic of birchbark letters. *Gap augmentation* (applied with probability 0.05) randomly replaces short runs of non-special tokens with the gap token #, training the model to restore characters in the presence of neighbouring unreadable sections — a situation common in heavily damaged documents. The aggregate masking rate across all three strategies is calibrated to remain at approximately 8%, consistent with the MLM setup.

3.5 Data Augmentation

For sampling balance purposes, all documents were divided into six thematic groups and assigned corresponding context tags: DAILY, LEGAL, CHURCH, LIT, EPIC, and SCIENCE. This grouping was not intended to be exhaustive or precise: assignments were based on the original characterisation of each source, and documents from sources covering multiple registers

were assigned to the most representative group. The tags were used solely to ensure a more balanced sampling distribution during training and do not constitute a linguistically motivated classification.

As shown in Fig. 5, religious texts are substantially overrepresented in the raw corpus. To address this imbalance, source-level weights were proposed for the training split: documents from underrepresented categories are sampled multiple times per epoch, while overrepresented sources are downweighted. The weights were calibrated to approximate the ordering $\text{DAILY} \approx \text{LEGAL} > \text{CHURCH} > \text{LIT} > \text{EPIC} > \text{SCIENCE}$, reflecting the thematic distribution of the target domain — birchbark manuscripts — in which everyday correspondence and legal documents constitute the majority of surviving texts.

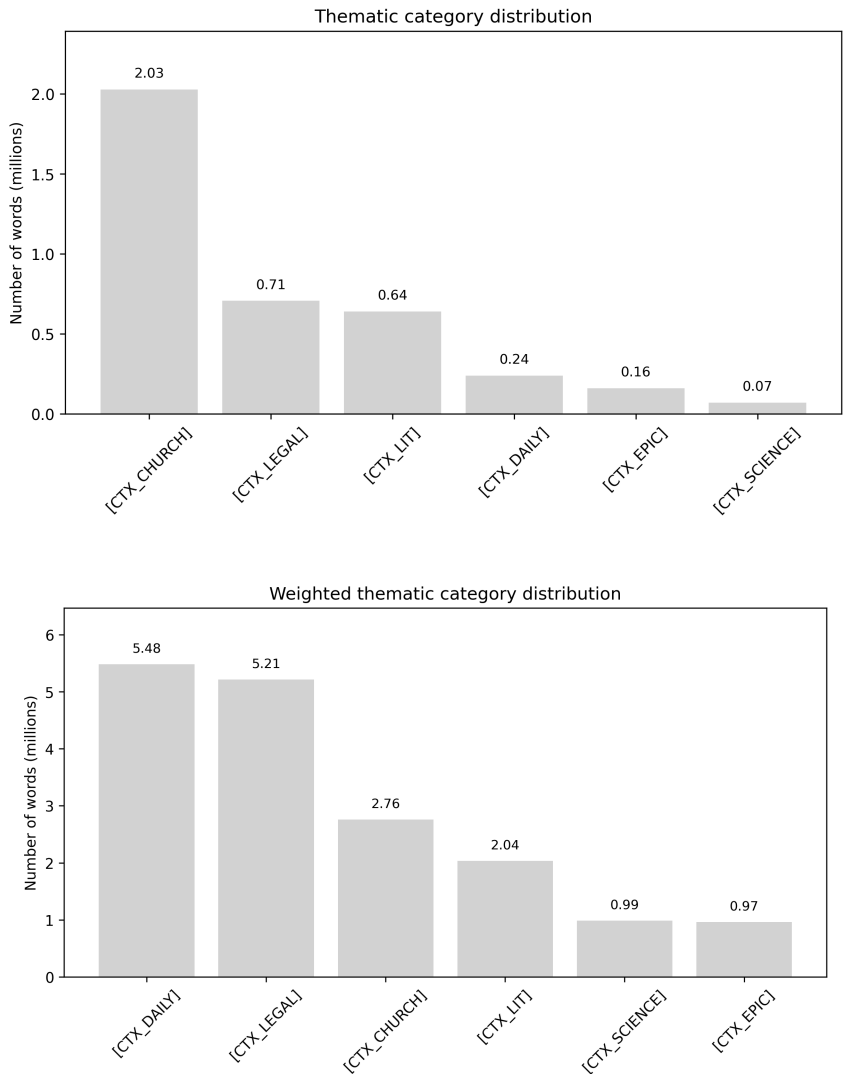


Fig. 7. Distribution of thematic categories in the dataset

4. Methods

A fundamental methodological question in computational lacuna restoration concerns the granularity of the prediction unit. Ithaca predicts sequences of individual characters via beam search, requiring the number of missing characters to be known in advance. Aeneas operates at the subword token level and additionally introduces a mechanism for inferring the number of missing tokens when it is unknown. The choice of granularity has consequences for both evaluation and generalisation: character-level prediction aligns naturally with the editorial markup of birchbark manuscripts, where lacunae are defined at the character level, while token-level prediction allows the model to leverage subword morphological structure, which may be advantageous for a highly inflected language.

Another significant issue is about whether restoration is performed position-by-position — predicting each masked unit independently — or jointly over the entire lacuna span via beam search. Independent per-position prediction is computationally cheap but ignores dependencies between the predicted units within a span; beam search is more expensive but allows the model to consider the joint probability of the restored sequence.

We evaluate two families of approaches along both dimensions discussed above. Fine-tuned models (Section 4.1) are evaluated in two modes: character-level per-position prediction, where outputs are restricted to single Cyrillic characters via constrained decoding, and token-level per-position prediction, where the full subword vocabulary is available. This dual evaluation allows us to isolate the contribution of prediction granularity from the contribution of domain fine-tuning, using the same model weights in both modes. From-scratch models (Section 4.2) are designed to support sequence-level beam search: RoFormer operates at the BPE token level, while DualEmbLM uses character-level tokenisation natively. The comparison between RoFormer and DualEmbLM, both trained from scratch on the same corpus, isolates the effect of tokenisation granularity on restoration quality, holding all other architectural and training choices constant.

4.1 Fine-tuned Models

Three pre-trained transformer encoders are selected for fine-tuning:

- mBERT (Devlin et al., 2018), [1] — a multilingual BERT model trained on 104 languages using a shared WordPiece vocabulary.
- BERTislav [2] — a ruBERT-based model pre-trained on a 10M-token corpus of Old Church Slavonic, Later Church Slavonic, Old East Slavic, Middle Russian, and Medieval Serbian texts.

- ModernBERT (Warner et al., 2024), [3] — a bidirectional encoder with an extended context window and rotary positional embeddings.

mBERT is included as a baseline because of its promising results on Akkadian lacuna restoration in Lazar et al. (2021). BERTislav is chosen because its pre-training domain partially overlaps with the target language variety. ModernBERT represents recent architectural improvements to the bidirectional encoder paradigm and serves as a more capable general-purpose baseline. All three models are fine-tuned on the assembled corpus using the MLM objective.

Table 1. Architectural comparison of three models.

Model	layers	attention heads	parameters	tokenization	Max seq length	Embeddings
mBERT	12	12	~179M	WordPiece	512	learned absolute
BERTislav	12	12	178M	WordPiece	512	learned absolute
ModernBERT	22	16	149M	BPE	8192	RoPE

Even after the preprocessing described in Section 3.3, a number of letterforms outside the standard Unicode Cyrillic block remain unresolved and fall outside the vocabularies of BERTislav and mBERT. For BERTislav, we apply the normalisation script provided by its authors, which maps non-standard forms to their standard equivalents (e.g., $\cdot \rightarrow \text{oy}$). For mBERT, we construct a custom normaliser derived from the model's vocabulary: characters absent from the tokeniser's vocabulary are mapped to their closest standard equivalents. ModernBERT, with its broader BPE vocabulary, does not require additional normalisation.

All three models are evaluated under constrained decoding: the prediction space is restricted to tokens corresponding to exactly one Cyrillic character (SingleCharCyrillicProcessor). This choice reflects a requirement of methodological consistency — since the restoration task is defined at the character level, all models, including subword-based ones, are evaluated in a unified output space. A consequence of this design is that zero-shot results for subword models reflect not only their linguistic competence but also the degree to which their pre-trained prediction space aligns with character-level granularity.

4.2 Models Trained from Scratch

4.2.1 RoFormer

The first from-scratch architecture is a RoFormer encoder (Su et al., 2021) trained from scratch on the assembled corpus using a BPE tokeniser with a vocabulary of 50,000 tokens. RoFormer replaces the absolute positional embeddings of standard BERT with rotary position embeddings applied to the query and key projections of each attention head, which has been shown to improve generalisation on longer sequences. The model follows a standard masked language modelling pre-training objective with the physical degradation collator described in Section 3.4. The architecture comprises 6 transformer layers, a hidden size of 512, 8 attention heads, and an intermediate feed-forward size of 2,048, yielding approximately 37M parameters in total.

Since the BPE tokeniser does not guarantee that token boundaries coincide with the boundaries of editorial reconstructions, masking on Test B is extended to the nearest token boundaries. As a result, tokens that partially overlap with the lacuna boundary are masked in full, causing the model to lose access to characters that lie outside the bracketed span but within the same token. This effect is deterministic and reproducible for a fixed tokeniser, and constitutes a mild disadvantage relative to the character-level models, which mask exactly the characters specified by the editorial markup.

4.2.2 DualEmbLM

The second from-scratch architecture addresses the granularity mismatch identified in Section 4.2.1 by operating directly at the character level. DualEmbLM uses a custom character vocabulary constructed from the training corpus and a word-level vocabulary of 50,000 entries, combining both via dual embeddings: each input position receives a concatenation of its character embedding and the embedding of the word to which it belongs, projected into the shared hidden space. This design draws on the intuition underlying Ithaca (Assael et al., 2022), where character-level representations are enriched with broader lexical context. The transformer encoder follows the same configuration as RoFormer: 6 layers, hidden size 512, 8 attention heads, intermediate size 2,048, trained with the physical degradation collator at a masking rate of 8% of characters.

Since each token corresponds to exactly one character, there are no tokenisation boundary effects at lacuna edges: every masked position corresponds to precisely one absent character, and the model predicts exactly one character per position. This alignment between the restoration target and the model's output unit is the key structural advantage over BPE-based approaches.

4.3 Auxiliary Classification

Building on the metadata described in Section 3.2, we train two auxiliary classifiers — for genre category and for conventional date — on top of the frozen embeddings produced by each of the five models. The classifiers are not trained end-to-end with the language models; instead, we extract fixed representations and train a lightweight probe on top of them.

For each model and each task, we train a linear probe: a single linear layer preceded by layer normalisation and dropout, applied to the pooled sequence representation. For BERT-based models the [CLS] token is used as the document representation; for RoFormer and DualEmbLM, the mean of non-padding token representations is used. Two versions of the input are evaluated: embeddings extracted from the restored text (*target*) and from the masked text with lacunae replaced by [GAP] tokens (*masked*), allowing us to measure the degradation in classification quality when the model operates on damaged rather than complete text.

Category classification is framed as a four-class problem (letters, records, religious, other) trained with weighted cross-entropy loss to address class imbalance (696 / 330 / 171 / 45 instances respectively). The probe is optimised with Adam (lr=1e-3) for 50 epochs; the best checkpoint is selected by macro-F1 on the validation set. Performance is reported as accuracy and macro-F1 on the test set, accompanied by a per-class confusion matrix.

Date classification follows the Aeneas approach: rather than predicting a single bin, the probe outputs a probability distribution over nine 50-year bins. The training target for each manuscript is a uniform distribution over the bins overlapping its conventional date interval, and the loss is KL divergence between the predicted and target distributions. The evaluation metric is the distance in years between the predicted mean date — the expected value of the predicted distribution — and the ground-truth interval, with a distance of zero awarded when the predicted mean falls within the interval.

All five models are evaluated on the same stratified train/val/test split (70/15/15), with stratification by category. Records with mixed-category annotations (containing //) are excluded from category classification; records without a parseable date interval are excluded from date classification.

5. Evaluation

5.1 Metrics

All models are evaluated on Test A and Test B using three complementary metrics. *Hit@k* measures whether the correct answer appears among the model's top-*k* predictions ($k \in \{1, 5, 20\}$); it is computed per character (fine-tuned and DualEmbLM) or per token (RoFormer),

and additionally at the span level for from-scratch models. *Character error rate (CER)* is the normalised Levenshtein distance between the predicted and target string. *Perplexity (PPL)* measures overall language modelling quality on the evaluation split. For from-scratch models, we additionally report *span-level macro-CER*, averaged uniformly across span lengths 1–10 following Aeneas; *span-level Hit@k* requires the entire reconstructed lacuna to match the target exactly.

The best checkpoint for each model is selected by Hit@1 on Test B in the corresponding prediction mode: character-level Hit@1 for fine-tuned models evaluated in character mode and for DualEmbLM, token-level Hit@1 for fine-tuned models evaluated in token mode and for RoFormer.

To reduce per-epoch evaluation time, the number of assessed gap positions is capped at 5,000 per test set. For token-level models, this limit is not reached nor for BERTislav and mBERT on both Test A (2,373–3,594 positions depending on tokenisation) and Test B (3,941 and 4,668 positions respectively); however, it is applied for ModernBERT on Test B. For character-level models, the limit is applied on both test sets for all three models.

5.2 Fine-tuned Models

5.2.1 Zero-shot

All three models are evaluated without fine-tuning under character-level constrained decoding. Performance is uniformly weak, which is expected: models originally trained to predict subword tokens perform poorly when restricted to single-character prediction. BERTislav achieves marginally higher results on Test A, consistent with its pre-training domain partially overlapping with the target.

Notably, on Test B Hit@1 scores are higher for all three models. ModernBERT achieves the highest Test B Hit@1 (0.069) and the lowest CER (0.911) among the three, indicating an architectural advantage even in the zero-shot setting.

Table 2. Zero-shot performance (characters).

Model	Test A		Test B	
	Hit@1	CER	Hit@1	CER
mBERT	0.014	0.915	0.057	0.942
BERTislav	0.017	0.911	0.057	0.930
ModernBERT	0.012	0.906	0.069	0.911

Table 3. Zero-shot performance (tokens).

Model	Test A		Test B	
	Hit@1	CER	Hit@1	CER
mBERT	0.014	0.915	0.057	0.942
BERTislav	0.017	0.911	0.057	0.930
ModernBERT	0.012	0.906	0.069	0.911

5.2.2 Fine-tuned

After five epochs of domain fine-tuning, all models improve (Table 4–5), however, the most prominent improvement is demonstrated by ModernBERT with Hit@1 score as twice as big as the others

Table 4. Fine-tune performance (characters).

Model	Best epoch	PPL	Test A			Test B		
			Hit@1	Hit@5	CER	Hit@1	Hit@5	CER
mBERT	4	4.24	0.020	0.068	0.897	0.079	0.315	0.906
BERTislav	2	8.11	0.022	0.071	0.916	0.074	0.274	0.915
ModernBERT	5	2.31	0.038	0.078	0.753	0.147	0.369	0.817

Table 5. Fine-tune performance (tokens).

Model	Best epoch	PPL	Test A			Test B		
			Hit@1	Hit@5	CER	Hit@1	Hit@5	CER
mBERT	4	4.15	0.761	0.877	0.180	0.271	0.452	0.687
BERTislav	5	7.24	0.702	0.822	0.253	0.204	0.340	0.779
ModernBERT	5	2.31	0.868	0.956	0.136	0.277	0.504	0.685

5.3 From-scratch Models

5.3.1 RoFormer

On Test A, the model achieves a token-level Hit@1 of 26.7% and Hit@20 of 49.6%, reflecting reasonable recovery of masked BPE units within a known-context setting. On Test B, performance drops substantially: token-level Hit@1 falls to 15.8%, and span-level Hit@1 — which requires the entire lacuna to be recovered exactly — reaches only 1.7%. The span-level macro-CER of 1.206 exceeds 1.0 because BPE tokens may decode to multi-character strings whose length differs from the bracketed target, inflating edit distance. This is a structural artefact of BPE tokenisation at lacuna boundaries rather than a failure of the language model per se.

Table 6. *RoFormer performance.*

Metric	Test A	Test B (token)	Test B (span)
PPL	131.25	8672.63	—
Hit@1	0.267	0.158	0.017
Hit@5	0.391	0.226	0.038
Hit@20	0.496	0.305	0.063
CER	0.839	0.982	1.206

5.3.2 DualEmbLM

The character-level architecture improves substantially over RoFormer across all metrics. On Test A, Hit@1 reaches 81.7% and CER 0.183. On Test B, character-level Hit@1 is 46.6% — nearly half of all individual lacuna characters are correctly recovered at rank 1 — and span-level Hit@1 is 22.2%. The span-level macro-CER of 0.589 remains below 1.0, in contrast to RoFormer, confirming that character-level tokenisation eliminates the BPE boundary artefact. The Test A–Test B gap reflects the increased difficulty of real lacunae: editorial reconstructions frequently occur in heavily damaged passages where adjacent context is itself uncertain. The combinatorial difficulty of span-level recovery — requiring every character to be correct simultaneously — further depresses span metrics relative to character-level ones.

Table 7. *DualEmbLM performance.*

Metric	Test A	Test B (char)	Test B (span)
PPL	4.26	16.24	—
Hit@1	0.817	0.466	0.222
Hit@5	0.916	0.676	0.282
Hit@20	0.983	0.909	0.328
CER	0.183	0.535	0.589

5.4 Auxiliary Classification

Tables 8–9 reports probing results for genre category (macro-F1, accuracy) and conventional date (mean and median distance in years) across all five models, evaluated on both restored (*target*) and damaged (*masked*) text.

Fine-tuned character models used for predictions

Table 8. Category prediction performance.

Model	Target dataset		Maked dataset	
	Acc	F1	Acc	F1
BERTislav	0.765	0.675	0.743	0.640
mBERT	0.770	0.661	0.727	0.633
ModernBERT	0.711	0.551	0.658	0.521
RoFormer	0.711	0.585	0.711	0.561
DualEmb	0.711	0.637	0.722	0.570

Table 9. Date prediction performance.

Model	Target dataset		Maked dataset	
	Mean dist	Median dist	Mean dist	Median dist
BERTislav	41.3	27.4	46.6	32.2
mBERT	45.6	34.3	47.7	35.6
ModernBERT	55.5	40.0	57.4	50.7
RoFormer	41.4	30.9	43.0	30.6
DualEmb	47.6	38.9	50.6	39.3

Several patterns emerge. First, BERTislav achieves the highest category macro-F1 on target text (0.675) and the best date mean distance (41.3 years), suggesting that its domain-specific pre-training encodes historically relevant structure more effectively than general-purpose multilingual models. Second, the target–masked gap is largest for ModernBERT in both tasks, indicating greater sensitivity to input degradation despite its superior restoration performance. Third, RoFormer shows minimal degradation between target and masked for category classification, which may reflect the robustness of BPE representations to partial character-level masking. DualEmb shows a modest accuracy gain on masked text for category, possibly because character-level representations are less affected by the substitution of lacuna content with [GAP] tokens than subword representations are.

5.5 HP-tuning

hyperparameters are selected on the validation set.

5.6 Experiment

Эксперимент с берестологом

6. Limitations & Future Work

6.1 Limitations

Masking rate mismatch. As discussed in Section 3.4, both token-level and character-level models are trained with a masking rate of 8%, chosen to match the empirical share of editorial reconstructions in Test B. However, this rate operates at different granularities: for character-level models, 8% of characters are masked, directly matching the target distribution; for token-level ones, 8% of BPE tokens are masked, each of which may span several characters, resulting in a substantially higher effective character-level masking rate. This discrepancy is an inherent limitation of applying a uniform masking rate across architectures

with different tokenisation granularities, and is acknowledged as such in the interpretation of results.

BPE boundary effects. When a lacuna boundary falls within a BPE token, the entire token is masked during Test B evaluation, causing the model to lose access to characters that lie outside the bracketed span but within the same token. This systematically widens the lacunae and constitutes a mild but consistent disadvantage for token-level models compared to character-level models, which mask exactly the characters specified by the editorial markup.

Dataset size. The birchbark and epigraphic components of the training corpus are relatively small (approximately 30,000 word tokens combined), and the models rely heavily on related but stylistically distinct sources to compensate. The degree to which this domain mismatch affects restoration quality on Old Novgorodian texts specifically remains an open question.

Normalisation coverage. The current normalisation pipeline does not handle digits, which remain in the training data as surface forms. Digit tokens are consequently included in the model vocabulary and may appear among top-k predictions for masked positions — a behaviour that is linguistically unmotivated for lacuna restoration. Future work should extend the normalisation to exclude or replace numeral forms.

6.2 Future Work

Several directions present themselves naturally. First, incorporating document-level thematic tags (e.g. LEGAL, CHURCH) as conditioning tokens — analogous to the CTX tags used in the data augmentation pipeline — could allow the model to leverage genre-level regularities in restoration. Second, extending the from-scratch architectures with a length prediction head, following Aeneas, would enable restoration in the more realistic setting where lacuna boundaries are uncertain. Third, evaluation on a larger and more balanced test set, stratified by lacuna length and document genre, would provide a more reliable basis for cross-model comparison.

7. Conclusions

This work presents a systematic comparison of four approaches to automatic lacuna restoration in Old Novgorodian birchbark manuscripts, spanning fine-tuned multilingual encoders and architectures trained from scratch on a purpose-built Old Russian and Old Church Slavonic corpus.

The principal finding concerns the from-scratch architectures. Character-level tokenisation substantially outperforms subword tokenisation in this setting: DualEmLM, which

combines character-level input with word-level context, achieves a Test B character Hit@1 of 46.6% and a span Hit@1 of 22.2%, compared to 15.8% and 1.7% respectively for RoFormer, which operates on BPE tokens. The BPE boundary artefact — whereby token boundaries do not align with lacuna boundaries — is identified as the primary structural cause of this gap, manifesting in a span macro-CER exceeding 1.0 for RoFormer.

Fine-tuned models evaluated in character-level mode perform substantially worse than either from-scratch architecture, with the best result of 0.147 for ModernBERT remaining well below DualEmbLM's 0.466. This reflects the mismatch between subword pre-training and character-level prediction that domain fine-tuning only partially alleviates. When the same fine-tuned models are evaluated in token-level mode — predicting subword units rather than individual characters — performance improves markedly: ModernBERT achieves a Test B token Hit@1 of 0.277, BERTislav 0.204, and mBERT 0.271. However, these results are not directly comparable to character-level metrics, as they are computed over different prediction units and different test set compositions. Within the token-level evaluation, ModernBERT consistently outperforms BERTislav and mBERT across all metrics and both test sets, confirming that recent architectural improvements to the bidirectional encoder paradigm transfer effectively to low-resource historical language modelling despite the absence of domain-specific pre-training.

The auxiliary probing experiments show that **all five models** encode genre and temporal information to a degree recoverable by a linear classifier. BERTislav achieves the highest category macro-F1, **consistent with its domain-specific pre-training**, while the target-masked degradation is most pronounced for ModernBERT, indicating greater sensitivity to input corruption despite its superior restoration performance.

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[1] <https://huggingface.co/npedrazzini/BERTislav>

[2] <https://huggingface.co/google-bert/bert-base-multilingual-cased>

[3] <https://huggingface.co/blog/modernbert>

Appendix 1. Dataset Information

Source	Language	Word tokens	Link
Birchbark manuscripts	Old Novgorodian (mostly)	21,464	gramoty.ru
Epigraphy	Old Church Slavonic (mostly)	8,102	epigraphica.ru
DIACU	Old Church Slavonic; Church Slavonic (Old Russian, Middle Bulgarian, Serbian, Resava recensions); Middle Russian	1,683,307	ACL Anthology
TOROT	Old Russian; Church Slavonic	682,430	torottreebank.github.io
Bible (Ponomar)	Church Slavonic	603,047	GitHub
Byliny	Old Russian (XI–XVII c.)	430,103	rusneb.ru
Pushkin House	Old Russian	256,503	lib2.pushkinskijdom.ru
Military Statute (Part 2)	Old Russian	49,787	rusneb.ru
NKRYA (historical)	Old Russian; Old Rus (XI–XVIII c.)	42,412	ruscorpora.ru
Total		3,777,155	

Appendix 2. Character Normalisation Mapping

The table below lists all substitutions applied during preprocessing. Characters marked Ø are deleted without replacement. Unlike conventional normalisation pipelines, historic Cyrillic letters (Ѡ, Ѣ, Ѥ, Ѧ, Ѩ, Ѭ, Ѯ, Ѱ, Ѳ) are preserved; only their variant encodings are unified.

Original	Unicode codepoint(s)	Replacement	Notes
<i>Cyrillic variant unification</i>			
Ѡ	U+046D+var	Ѡ	ot-ligature variant
ѡ	U+1F7C	Ѡ	Greek omega used as ot-ligature
Ѣ	U+A64D	Ѣ	omega variant
ѣ	U+047B	ѣ	round omega
Ѥ	U+A66B	Ѥ	omega with two eyes → omega with eye
ѥ	U+A66D	Ѥ	omega with iotified eyes → omega with eye
Ѧ	U+A645	Ѧ	reversed dze
ѧ	U+1D24	Ѧ	dze variant
Ѩ	U+0225	Ѩ	dze variant
ѩ	U+0223	Ѥ	ou-ligature
Ѫ	U+A665	Ѫ	l variant
ѫ	U+A647	ѫ	iota
Ѭ	U+2C5A	Ѱ	fita variant
ѭ	U+A65B	Ѩ	big yus variant
Ѯ	U+A659	Ѯ	little yus variant
ѯ	U+048D	ѯ	yat variant
Ѱ	U+2C55	Ѱ	ye variant
ѱ	U+2C54	Ѱ	ye variant
Ѳ	U+A655	Ѳ	reversed yu
ѳ	U+2C34	ѳ	d-variant (Glagolitic-derived)
Ѵ	U+2C49	Ѡ	ot-ligature variant (Glagolitic-derived)
ѵ	U+A649	ѳ	g-variant (Glagolitic-derived)
Ѷ	U+2C39	Ѷ	i-variant (Glagolitic-derived)
ѷ	U+2C56	Ѩ	iotified big yus variant (Glagolitic-derived)
Ѹ	U+1E2F	Ѷ	i with diaeresis and acute
ѹ	U+0477	ѹ	izhitsa variant
Ѻ	U+04D3	Ѻ	a with diaeresis
ѻ	U+04C5	ѻ	i with macron
Ѽ	U+04F1	Ѽ	u with diaeresis
ѽ	U+04F9	ѽ	y with diaeresis
<i>Titlo and abbreviation marks</i>			

Original	Unicode codepoint(s)	Replacement	Notes
˘	U+0487	˘	combining titlo variant → canonical titlo (U+0483)
˙	U+1D55	˘	modifier letter combining ring → titlo
Punctuation			
†	U+2020	+	dagger → cross
×	U+00D7	+	multiplication sign → cross
⋮	U+2058	:	four-dot punctuation
⋮	U+2059	:	five-dot punctuation
⋮	U+205E	:	four dots vertical
	U+00A6	:	broken bar
·	U+2219	·	bullet operator → middle dot
*	U+002A	·	asterisk → middle dot
.	U+002E	·	full stop → middle dot
Private Use Area			
□	U+F13F	·	PUA separator → middle dot
□, □, □, □, 喀	U+F222, U+F23A, U+F2B4, U+F2B5, U+F4A5	˘	PUA titlo variants
⌘	U+F074	s	PUA dze variant
□, 穰	U+F130, U+F48E	·	PUA omega with eye variants
□	U+F147	w	PUA omega variant
□, □	U+F14E, U+F42E	ŵ	PUA ot-ligature variants
□	U+F467	Ꜥ	PUA xi variant
詭, 𐄂	U+F47E, U+F480	·	PUA uk variants
Deleted characters (∅)			
˘˘˘	U+0307, U+0308, U+0334, U+0358	∅	combining diacritics
˘	U+2E2F	∅	wavy hyphen
·	U+A63F	∅	combining titlo (decoration)
˘˘	U+02B9, U+02BC	∅	modifier letters
ꝛ	U+0481	∅	combining pen flourish (koppa)
○ []	U+20DD, U+27E6, U+27E7	∅	enclosing marks
/ \ ? ! " ' ; ,	U+002F, U+005C, U+007C, U+003F, U+0021, U+0022, U+003B, U+002C	∅	punctuation not used in corpus
(LRM)	U+200E	∅	left-to-right mark
𐄂 □ □ 𐄂 □ 𐄂	U+F080, U+F245, U+F265, U+F27A, U+F2DB, U+F4A4	∅	PUA glyphs